

KUSHAGRA AGGARWAL

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PROFESSIONAL SUMMARY:

AI/ML Engineer and MS Data Science student at Northeastern University with 3+ years shipping production-grade LLM agents, RAG pipelines, and ML systems using Python, PyTorch, LangChain, and AWS. AWS Certified Data Engineer Associate.

EDUCATION:

Northeastern University: - Master of Science in Data Science | Expected May 2027

Certifications: - **AWS Certified Data Engineer Associate** (Oct 2025 – Oct 2028)

TECHNICAL SKILLS:

- **AI/ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, LangChain, OpenAI APIs
- **LLMs & Agents:** RAG Pipelines, Multi-Agent Orchestration, Prompt Engineering, Vector DBs (FAISS, Chroma, Pinecone), Fine-tuning (LoRA/QLoRA), Semantic Embeddings
- **MLOps & Deployment:** Docker, Git/GitHub, CI/CD, Model Serving, Inference Optimization, Autonomous Workflows, Terraform (exposure)
- **Programming:** Python (pandas, NumPy, SciPy), SQL, R, PySpark, Bash
- **Cloud & Data:** AWS, GCP, Azure, Snowflake, Databricks, ETL Pipelines, Azure Data Factory
- **Tools:** Jupyter, VS Code, Power BI, Tableau, Anaconda

EXPERIENCE:

Questwalk Technologies Pvt Ltd, India | Data Scientist | Nov 2023 – Nov 2024

- Architected end-to-end ML pipelines on AWS/GCP using PyTorch and Scikit-learn, cutting manual workflow effort by **50%** and lifting F1 scores by **18%**.
- Deployed LLM-powered agents with LangChain, OpenAI APIs, and FAISS vector store, automating workflows that previously consumed 8+ analyst hours per week.
- Engineered RAG pipelines with semantic embeddings and context injection, reducing hallucination rates by **35%** across production AI applications.
- Optimized model training on cloud GPU environments through hardware-software profiling, accelerating inference throughput by **40%**.
- Created Power BI dashboards on Snowflake delivering real-time KPI visibility, slashing ad-hoc reporting requests by **30%**.
- Collaborated with engineering teams on production model integration, applying Git version control, code review, and release management across **5+** projects.

Techminds IT Services Pvt Ltd, India | Data Analyst Intern | May 2022 – Jul 2023

- Built Algorithmic ETL pipelines on AWS/GCP with Python and SQL, improving processing reliability and reducing manual data prep by **60%**.
- Developed NLP preprocessing pipelines for ML model consumption, structuring 100K+ unstructured text records into embedding-ready formats.
- Maintained ML models and pipelines in Git repositories with modular design across **15+** collaborative engineering workflows.
- Tuned SQL queries and ETL logic alongside senior engineers, boosting data throughput by **25%** across production ML pipelines.

Stanvac System, India | Data Analyst Intern | Jan 2021 – Feb 2022

- Implemented Python-based data processing workflows producing model-ready datasets for **10+** downstream ML pipelines.

- Automatic repetitive data extraction tasks with Python scripts, reducing manual workload by **45%** across recurring operations.
- Documented pipeline logic and configurations in Git, enabling reproducibility across collaborative engineering projects.

PROJECTS:

Healthcare Question Answering with Large Language Models ([Github](#)) Python · PyTorch · LangChain · RAG · OpenAI APIs · Statistical Evaluation

- Shipped production-grade RAG agent with FAISS vector store and transformer models, delivering grounded AI responses across **10K+** medical documents.
- Engineered automated ingestion pipeline converting unstructured text into embedding-indexed formats, achieving **92%** retrieval relevance.
- Benchmarked LLM configurations on hallucination, latency, and cost, iterating prompts to cut hallucinations by **40%**.

AI-Powered Adaptive Traffic Signal Control Using Reinforcement Learning ([Github](#)) Python · DQN/PPO · TensorFlow · PyTorch · Statistical Analysis · Experimental Design

- Designed deep RL system (DQN/PPO) optimizing real-time traffic flow, reducing average vehicle wait time by **38%** vs. static baselines.
- Built Programmed data pipelines using SUMO/VISSIM simulation environments for reproducible RL training across **6** model configurations.
- Profiled training performance across GPU configurations, accelerating training throughput by **30%** through architecture optimization.